POGO PINS

Comprehensive Installation & Design Guide

Topics Covered:
SMD & Through-Hole Installation | PCB Footprint Design | Alignment Fixtures | Critical Parameters

Reference Guide for Engineers & PCB Designers

1. WHAT ARE POGO PINS?

A **Pogo pin** (spring-loaded contact pin) is a precision electrical connector consisting of three core parts: a **plunger** (contact tip), an internal **spring**, and a **barrel/tube** (housing). When compressed against a mating surface, the spring maintains consistent contact force, ensuring reliable electrical continuity even with minor height variations or misalignment.

They are widely used in test fixtures, charging docks, programming jigs, board-to-board connectors, and RF probing. Common tip styles include crown, spear, cup, cone, serrated, and dome — each optimized for different contact surfaces (flat pads, through-hole pins, contaminated surfaces, etc.).

Key Applications:

- ICT (In-Circuit Test) & FCT (Functional Circuit Test) fixtures
- Battery charging contacts (smartwatches, earbuds, medical devices)
- Programming & debug interfaces
- Board-to-board modular connections
- RF & high-frequency test probes

Note: *Pogo pins are rated by working stroke, spring force, current capacity, and durability (cycle life). Always select a pin rated for your application's mechanical and electrical requirements.*

2. SURFACE MOUNT (SMD) INSTALLATION

SMD Pogo pins are the most common choice for modern consumer electronics because they save space and are compatible with automated pick-and-place and reflow soldering. However, they require careful footprint design and process control.

2.1 PCB Footprint Design

Pad Diameter:

Scale the pad diameter with the pin's barrel diameter. A slightly larger pad than the barrel base creates a strong solder fillet and improves both mechanical strength and electrical continuity. Too small a pad leads to weak joints; too large invites bridging in dense arrays.

Pad Shape — CRITICAL:

Use **circular pads** rather than elongated or oval ones. Because Pogo pins carry axial spring load, an oval pad allows the pin to twist under load, changing the contact angle and causing intermittent connections or solder joint fatigue. A circular pad centers the part and resists rotation.

Soldermask Clearance:

Allow slightly wider soldermask clearance than standard passive components. Pogo pin pads carry more solder paste volume, increasing the risk of solder bridging in dense arrays.

Keepout Zone:

Account for the pin's compressed spring envelope and housing clearance. In multi-pin arrays, verify that adjacent pins don't mechanically interfere when fully compressed. Check the datasheet for the pin's maximum compressed diameter.

2.2 Solder Paste & Stencil

Use **fine-tuned stencil apertures**. Too much solder paste can cause **solder wicking** — molten solder flows into the pin barrel during reflow and permanently jams the plunger, killing the spring action. This is one of the most common failure modes in SMD Pogo pin assembly.

Stencil thickness and aperture design must be carefully matched to the pin base size. Consider using a stepped-down stencil (thinner in the Pogo pin region) if the rest of the board requires standard thickness.

2.3 Placement

Place with a pick-and-place machine using a vacuum nozzle that matches the Pogo pin shape to avoid damaging the cap, plunger, or barrel. The pin must sit perfectly flat on the solder pad and remain aligned with the mating contact area.

2.4 Reflow Soldering

Use a standard lead-free reflow profile, but monitor peak temperature and exposure time carefully. Excessive heat can damage the internal spring (annealing) or degrade the gold plating. Reflow-rated Pogo pins can run through the oven in the same pass as other SMT components. Non-reflow-rated pins must be hand-attached afterward.

2.5 Post-Reflow Inspection

Inspect for tombstoning, misalignment, and solder bridges. Clean flux residue without affecting the spring mechanism. Verify that each pin's plunger moves freely and returns to full extension.

> ***Warning:*** *Never force-compress a Pogo pin before the solder has fully cooled and solidified. Doing so can crack the solder joint or shift the pin position.*

3. THROUGH-HOLE (TH / DIP) INSTALLATION

Through-hole mounting is preferred for applications requiring higher mechanical strength, such as industrial equipment, power connections, and products exposed to vibration or frequent mating cycles.

3.1 PCB Hole Design

Hole Diameter:

Start with the pin tail diameter and add clearance for solder fill. Too tight, and solder can't wick through; too loose, and the pin loses mechanical grip and may tilt under repeated compression cycles. A good rule of thumb is tail diameter + 0.1–0.15 mm for standard pins.

Annular Ring:

Follow IPC-2221 minimums to ensure the plated-through connection remains reliable after accounting for drilling tolerance and registration errors. For critical applications, use larger annular rings to improve joint strength.

3.2 Assembly Process

1. **Insert** the Pogo pin tail through the drilled hole from the component side.
2. **For multi-pin connectors**, use a plastic housing or carrier to hold each pin in position. This maintains correct pitch and prevents pin tilt during soldering.
3. **Solder** on the opposite side using a temperature-controlled soldering iron. Apply heat to both the pad and the pin tail simultaneously.
4. **Use high-quality flux** to improve solder flow and joint strength. Avoid cold joints.
5. **Clean** flux residue carefully after soldering — some residues become corrosive over time or interfere with electrical contact.

3.3 Mechanical Considerations

Through-hole Pogo pins create a strong mechanical anchor that handles more stress than SMT designs. The tradeoff is increased assembly complexity (drilling + soldering) and higher PCB space usage. For very high-cycle applications, consider using **receptacles** (sockets) that hold the Pogo pin and allow easy field replacement without desoldering.

> ***Tip:*** *For hand-soldering prototypes, pre-tin both the pin tail and the PCB pad before insertion. This ensures better wetting and a stronger joint with minimal heat exposure.*

4. CRITICAL DESIGN PARAMETERS

The following parameters must be verified for both SMD and TH installations to ensure reliable operation over the product lifetime.

Parameter	Description	Design Guideline
Working Stroke	The designed compression distance for reliable contact	Design to working stroke (not total travel). Check min/max tolerance stack
Board-to-Board Gap	Distance between mating surfaces when connected	Must match pin working stroke across full tolerance stack (housing, PCB t
Mating Surface	The contact pad or pin on the mating side	Must be clean, flat, and plated (gold preferred) for low contact resistance.
Placement Tolerance	Positional accuracy of the pin on the PCB	SMT pins are sensitive to rotation. Build angular tolerance into assembly r
Current & Thermal	Electrical load and heat dissipation	For high-current pins, provide adequate copper area and thermal vias at t
Spring Force	Contact pressure exerted by the spring	Too low = poor contact; too high = excessive wear on mating surface. Mat
Cycle Life	Number of compression cycles before failure	Consumer: 10K–30K; Industrial/Test: 100K+. Select pin rated for expected

4.1 Working Stroke — The Most Critical Parameter

Working stroke is the compression distance the pin is designed to operate within. Too little compression results in unstable contact and high resistance; too much compression damages the spring or causes permanent set (the spring loses its ability to return). Always design to the working stroke, not the total travel, and verify against minimum/maximum tolerance extremes — not just the nominal dimension.

4.2 Tolerance Stack-Up Analysis

For board-to-board connections, the gap between mating surfaces must stay within the pin's working stroke across the entire tolerance stack. This includes: PCB thickness variation, component height tolerance, housing dimensional tolerance, and solder joint height. A gap at the minimum tolerance extreme may over-compress the pin; at the maximum extreme, the pin may not make contact at all.

5. WHEN ARE ALIGNMENT FIXTURES REQUIRED?

An **alignment fixture** (or test jig) becomes essential in scenarios where manual alignment is insufficient or where multiple pins must contact simultaneously with high precision.

5.1 Multi-Pin Test Fixtures & Bed-of-Nails

When probing dozens or hundreds of test points simultaneously, a fixture is mandatory. A typical Pogo pin test fixture consists of:

Pogo pins (test probes) — spring-loaded contacts selected for the target test point type

Receptacles — sockets that hold pins and allow easy replacement without re-soldering

Probe plate — a precision-machined plate (often Delrin or aluminum) that holds pins in exact alignment with PCB test points

Test Point Carrier Board (TPCB) — routes signals from the Pogo pins to the test equipment

Fixture base — provides structural integrity, guiding mechanisms, and clamping force

5.2 High-Density or Fine-Pitch Applications

As pitch decreases (e.g., JST ZH at 1.5 mm or smaller), the cup size of the Pogo pin must be large enough to accept the target pin but small enough to avoid clearance issues with neighboring pins. Board-to-fixture alignment becomes critical. At fine pitches, even a 0.1 mm misalignment can cause probes to miss their targets or short adjacent pins.

5.3 Tight Tolerance Stack-Up

If the cumulative tolerance of connector position + tooling hole tolerance + Pogo pin position must stay under approximately 0.5 mm, a precision fixture is required. Relying on board outline alone is insufficient. **Always use tooling holes in the PCB** for alignment — these provide far more repeatable positioning than board edges, which have larger manufacturing tolerances.

5.4 Production Testing & Programming Jigs

For repeatable, high-volume testing, fixtures ensure:

- Consistent compression force across all pins (no weak contacts)
- Proper DUT (Device Under Test) seating before applying power
- Protection against damage from misalignment or skewed insertion
- Rapid changeover between different board variants using interchangeable probe plates

5.5 Clamping & Support Requirements

Even though each pin may only exert a few grams of force, a 100-pin fixture requires substantial total clamping force (potentially several kilograms). The DUT must be **well supported** to prevent PCB bending, which can cause intermittent contact or permanent board damage. Toggle clamps are commonly used, but they need a mechanical interface with the DUT that distributes force evenly.

5.6 Detection / Safety Pins

Add 2–4 extra Pogo pins set slightly lower than the main array, connected to ground on the DUT. When these make contact, the fixture confirms the board is fully seated and can safely apply power. This prevents applying voltage to partially seated boards, which could damage the DUT or the fixture.

> ***Fixture Design Best Practice:*** *Design the fixture so that the Pogo pins are replaceable. Use receptacles instead of direct-soldering pins to the TPCB. Pogo pins are wear items and will need replacement over time.*

6. SMD vs. THROUGH-HOLE COMPARISON

Aspect	SMD Pogo Pins	TH Pogo Pins
Best For	Consumer electronics, high-density boards	Industrial, power, high-vibration environments
Mechanical Strength	Lower (relies on solder joint)	Higher (strong mechanical anchor through PCB)
Assembly Method	Pick-and-place + reflow soldering	Drill + insert + solder (manual or wave)
PCB Space Usage	Compact footprint	Requires more PCB area (holes + pads)
Rework / Repair	More difficult; risk of pad damage	Easier, especially with receptacles
Alignment Aid	Circular pads, precise stencil design	Plastic housing/carrier for multi-pin arrays
Cost (Assembly)	Lower (automated)	Higher (additional process steps)
Thermal Performance	Limited (solder joint is heat path)	Better (pin tail conducts heat through board)
Vibration Resistance	Moderate	Excellent

7. KEY TAKEAWAYS

1. Choose the right mounting method for your mechanical and environmental requirements. SMD for density and automation; TH for strength and reliability.

2. Control solder paste volume religiously for SMD pins to prevent barrel wicking — the #1 cause of SMD Pogo pin failure.

3. Design PCB footprints around the specific pin — pad/hole size, shape (circular for SMD!), and keepout all matter.

4. Always check tolerance stack-up at min/max extremes, not just nominal, for compression gap and mating surface alignment.

5. Use alignment fixtures whenever you have multi-pin arrays, fine pitches, tight tolerances, or production test requirements. Always align using tooling holes, not board edges.

6. Specify working stroke, not total travel in your design. Total travel includes over-compression margin; working stroke is the reliable operating range.

7. Consider receptacles for high-cycle or field-serviceable applications. They make Pogo pin replacement fast and solder-free.

— End of Guide —